

TBA4245 GEODESY**Example exercise no. 12: Gross error test, external reliability. Points.**

Solution included.

Example of gross error test and external reliability test. Points.

Four property boundary points ("grensepunkt") are measured in with GNSS real time measurements. The network of control points (reference stations) in CPOS, owned by the Norwegian mapping authority ("Kartverket") are used. Only the coordinates are tested, not the heights. Normally the heights are of no interest for border points. The exception is three dimensional border points.

In the following example, four border points on the figure are measured at least two times. The points C1 and C2 are measured three times, C3 and C4 twice. The numeric values in the example are calculated with the program GISLINE. In the example, there are two to four hours between the rounds of the measuring-in.

In the guidelines in the Norwegian standard "Satellite-based geo referencing" «Satellittbasert posisjonsbestemmelse» it is recommended that there should be at least 45 minutes between two measurements of the same point to obtain independent measurements. Alternatively, three measurements are performed by at least 15 minutes interval. It is recommended to turn the equipment on and off completely between each measurement.

The measured data are the coordinates and heights. You must choose to export the measurements as point observations. Table 2 shows observation data for point C1-round 2 in the export file (KOF-file). The coordinates and height of a point is shown in line 50; NB: Ellipsoidal heights. In line 47 the numerical values of the variance-covariance matrix of X, Y and Z are shown (the Cartesian coordinates X, Y, Z). This information is needed for the analysis of accuracy. In a land surveying program, we can, from the values of the accuracy measures in line 47, calculate the standard deviation of measured values in the map projection plane: *N*, *E* and *H*, and the correlation coefficients. See table 2.

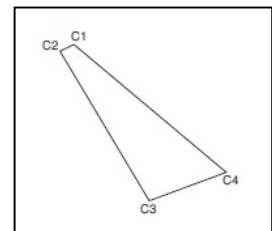
Table 1 Example of a KOF file for a point observation (punktobservasjon)

```
-46 DDDMMYYYY HH:MM:SS SV DD.DD II.III PPP MMM
-50 P P P P P P P P P P K K K K K K K K X X X X X X X X . X X X Y Y Y Y Y Y Y Y . Y Y Y E L L H . H H H B k M M M M M M M
-47 v v v . X X X X X X X X v v v . Y Y Y Y Y Y Y Y v v v . Z Z Z Z Z Z Z Z v v v . X Y X Y X Y X Y v v v . X Z X Z X Z X Z v v v . Y Z Y Z Y Z Y Z
50 C1 1 7100907.536 622902.793 111.177
47 0.00003383 0.00001662 0.00007647 0.00000901 0.00000938 -0.00000342
46 07072013 14:39:51 1.3 0.000
```

Table 2 Gisline, observations and accuracy measures. Two parts, see also next page.

Point, Mnr: Målnummer, Round no., Inr: Instrument no., Coordinates (UTM), Height (NN2000), Antenna height

	Punkt	Mnr	Tema	Inr	Type	X	Y	H	Ant.h
	C1	1 {PUNKTOBS}		61	P	7100907.53200	622902.774	74.19462	0.000
	C1	2 {PUNKTOBS}		61	P	7100907.53600	622902.793	74.22062	0.000
*	C1	3 {PUNKTOBS}		61	P	7100907.4580	622902.796	74.1826	0.000
*	C1	4 {PUNKTOBS}		61	P	7100907.5170	622902.776	74.1496	0.000
	C2	1 {PUNKTOBS}		61	P	7100906.3270	622899.659	73.7204	0.000
	C2	2 {PUNKTOBS}		61	P	7100906.3290	622899.660	73.7634	0.000
	C2	3 {PUNKTOBS}		61	P	7100906.3380	622899.664	73.7124	0.000
*	C2	4 {PUNKTOBS}		61	P	7100906.2910	622899.651	73.7345	0.000
	C3	1 {PUNKTOBS}		61	P	7100876.0180	622917.954	73.2812	0.000
	C3	2 {PUNKTOBS}		61	P	7100876.0100	622917.934	73.3092	0.000
*	C3	3 {PUNKTOBS}		61	P	7100876.0240	622917.953	73.2772	0.000
*	C3	4 {PUNKTOBS}		61	P	7100876.0280	622917.946	73.2952	0.000
	C4	1 {PUNKTOBS}		61	P	7100881.6250	622933.835	76.0531	0.000
	C4	2 {PUNKTOBS}		61	P	7100881.6380	622933.822	76.0871	0.000
*	C4	3 {PUNKTOBS}		61	P	7100881.6370	622933.839	76.0481	0.000
*	C4	4 {PUNKTOBS}		61	P	7100881.6180	622933.829	76.0491	0.000



sX etc.: Standard deviations on X, Y and H, rXY etc.: Correlation coefficient between X and Y etc., Date, Time, PDOP

sX	sY	sH	rXY	rXH	rYH	Dato	KI	PDOP
0.00445	0.003	0.010	-0.156	0.396	-0.165	20130707	12:57:26	1.300
0.00613	0.004	0.009	-0.295	0.196	-0.079	20130707	14:39:51	1.300
0.0078	0.004	0.014	-0.424	0.758	-0.496	20130707	18:16:52	1.600
0.0037	0.003	0.010	-0.013	0.161	0.247	20130707	21:12:21	1.300
0.0074	0.005	0.016	-0.377	0.280	-0.132	20130707	12:56:05	1.300
0.0068	0.004	0.009	-0.146	0.286	-0.121	20130707	14:38:56	1.300
0.0083	0.004	0.013	-0.350	0.728	-0.505	20130707	18:15:27	1.700
0.0042	0.003	0.010	0.087	0.247	0.203	20130707	21:11:02	1.300
0.0099	0.008	0.018	-0.370	0.328	-0.110	20130707	12:53:15	1.400
0.0076	0.004	0.010	0.020	0.164	-0.097	20130707	14:37:25	1.300
0.0056	0.003	0.011	-0.017	0.619	-0.305	20130707	18:13:43	1.300
0.0039	0.003	0.012	-0.052	0.144	0.299	20130707	21:08:58	1.800
0.0061	0.004	0.013	-0.361	0.276	-0.206	20130707	12:52:02	1.400
0.0075	0.005	0.013	0.058	0.147	0.031	20130707	14:36:11	1.400
0.0085	0.005	0.019	-0.195	0.593	-0.530	20130707	18:11:56	1.700
0.0038	0.003	0.010	0.046	0.286	0.257	20130707	21:07:50	1.500

The measured values are found in the observation file in Gisline, see the table 2 above (is split into two tables).

NB: The observations marked with an asterisk in column no. 1 are not used (they are set as passive obs.). But, observation C1-3 was set passive after the gross error test.

Transformation of the accuracy measures

Relevant theory in the text book "Basic GPS".

In table no. 1 the measured data of observation no. 2 of point C1 are shown. In table no. 2 this point observation is named C1 with Mnr (round number) equal 2 (second line), C1-2.

Measured coordinates of C1-2:

Table 3 Coordinates

	North	East
UTM coordinates	N = 7100907,536 m	E = 622902,793 m
Longitude, latitude	64,0131803° N	11,51440559° E Greenwich

Transform the variance-covariance matrix of Cartesian coordinates X, Y and Z found in table 1, to the variance-covariance matrix of the UTM coordinates and the NN2000 height.

Calculate the standard deviations of the UTM coordinates and the NN2000 height.

Calculate the three correlation coefficients between the UTM coordinates and the NN2000 height.

Check your results by comparing with the values from table 2: sX, sY, sH, and rXY, rXZ and rYZ.

Gross error test. Grovfeilanalyse (observasjonstest)

Relevant theory in the text book "Reliability analyses of networks of control points".

Normally we start with a gross error test where all the point observations are used (i.e: 3 point observations of C1 and C2, and 2 point observations of C3 and C4).

Table 4 Printout from Gisline, the report of the gross error test:

UTJEVNING I GRUNNRIS (estimation in the plane, 2D)						
FRI UTJEVNING (free estimation, free network)						
TEST AV OBSERVASJONER - MULTIPPEL T-TEST						
Fra	From	Til	To	Restfeil Residual	Est.grovfeil Estimated gross error	Testverdi Test value

C1		1	X	-0.011	0.019	1.00
C1		1	Y	0.010	-0.014	1.12

C1	2	X	-0.015	0.028	1.37
C1	2	Y	-0.009	0.017	1.39
C1	3	X	0.063	-0.067	4.46 *
C1	3	Y	-0.012	-0.003	0.18
C2	1	X	0.004	-0.008	0.29
C2	1	Y	0.003	-0.005	0.26
C2	2	X	0.002	-0.003	0.12
C2	2	Y	0.002	-0.003	0.19
C2	3	X	-0.007	0.013	0.44
C2	3	Y	-0.002	0.006	0.40
C3	1	X	-0.002	0.015	0.40
C3	1	Y	-0.015	0.021	0.80
C3	2	X	0.006	-0.015	0.40
C3	2	Y	0.005	-0.021	0.80
C4	1	X	0.007	-0.011	0.36
C4	1	Y	-0.007	0.012	0.59
C4	2	X	-0.006	0.011	0.36
C4	2	Y	0.006	-0.012	0.59

Tabellverdi (Tabulated value)=3.88 (Student-t, f=11, alfa=0.0013)

OPPSUMERING ETTER TEST AV OBSERVASJONER (Summary):

Kategori	Ant.obs.	Akkumulert (%)

Test/Tabell < 1.0	19	95.00
1.0 < Test/Tabell < 2.0	1	100.00
2.0 < Test/Tabell < 3.0	0	100.00
3.0 < Test/Tabell < 3.0	0	100.00
Ukontrollerbar	0	100.00

Parameters of the multiple Student T test: Compare your calculations with values in the printout above:

Show that the degree of freedom, f , of the test is 11.

Show that the confidence level, α , of the test is $0.0013 = 0.13\%$.

Check that the tabulated value of the test is 3.88.

Results of the gross error test:

Explain *residual* and *estimated gross error* in the table above.

Comment the result of the gross error test. Are there any gross error(s)?

Probably you found a gross error. Describe the procedure after gross error(s) is/are detected.

External reliability. «Ytre pålitelighet»

Table 5 The result of the external reliability test

YTRE PÅLITELIGHET		External reliability				
DATUM: EUREF89 - SONE 32						
UTJEVNING I GRUNNRIS		Estimation in 2D				
TVUNGEN UTJEVNING		Constraint estimation				
NYBESTEMTE KOORDINATER MED MIDLERE FEIL (standardavvik) [meter]						
Estimated coordinates and their standard deviations						
PUNKT (point)	N	E	H	sN	sE	sH

C1	7100907.534	622902.783		0.006	0.004	
C2	7100906.331	622899.662		0.008	0.004	
C3	7100876.016	622917.939		0.010	0.006	
C4	7100881.632	622933.828		0.008	0.006	

YTRE PÅLITELIGHET - KOORDINATER [meter]					
External reliability - Coordinates					
KOORDINAT	Observasjon....[meter/gon]			Indre pål.	Ytre pål.
Coordinate	Observation,no.			Internal reli.	External rel.

P C1	C1	1	X	-0.041	0.027
P C2	C2	1	X	-0.043	0.015
P C3	C3	2	Y	-0.054	0.043
P C4	C4	1	X	-0.050	0.031
Normparametre:					
Normtype	: Stedfesting av matrikkelenhets- og råderettsgrenser				
Norm type	: Geo referencing of land register- and right-of-use borders				
Normklasse	: 1 : Byområde (høy grad av utnyttning) (k = 100 mm)				
Norm class	: 1 : Urban area, high degree of utilization (k = 100 mm)				
Tabellverdi = 2.26 (Student-t, f = 9, alfa = 0.0250)					
Tabulated value = 2.26 (Student-t, f = 9, alpha = 0.0250)					

Parameters of the external reliability test: Compare your calculations with values in the printout above:

Show that the degree of freedom, f , of the test is 9.

Why is the confidence level (significance level), α , of the test $0,0250 = 2,5\%$.

Check that the tabulated Student-T value of the test is 2,26.

Explain the actual chosen “Norm class”

Results of the external reliability test:

Why do we prefer to use a constraint network for external reliability analysis?

In this example, is it a constrained network?

Explain the values of the internal reliability (“indre nøyaktighet”) in the table above.

Comment the result of the external reliability test. Are the points Ok?

Why does point C2 have a much lower numerical value of the external reliability compared to the other three points, 0,015 m vs 0,027/0,043/0,031 m?

For point C2: Explain in words what causes the external reliability value of 0,015 m.

Compare the external reliability values of points C1 and C2 with the standard deviations of the estimated coordinates shown in the table above.

Solution

See also in the text book “Grunnleggende landmåling” where this example is used in chapter 10, page 245 ff (in the 2014 edition).

Transformation of the accuracy measures

Relevant theory in the text book “Basic GPS”.

Transform the variance-covariance matrix of Cartesian coordinates X, Y and Z found in table 1, to the variance-covariance matrix of the UTM coordinates and the NN2000 height.

From the KOF file, the variance-covariance matrix of Cartesian coordinates X, Y and Z, C_{XYZ} , in m^2 and mm^2 :

$$C_{XYZ} = C_{CTRS} = \begin{bmatrix} 0,00003383 & 0,00000901 & 0,00000938 \\ 0,00000901 & 0,00001662 & -0,00000342 \\ 0,00000938 & -0,00000342 & 0,00007647 \end{bmatrix} m^2 = \begin{bmatrix} 33,83 & 9,01 & 9,38 \\ 9,01 & 16,62 & -3,42 \\ 9,38 & -3,42 & 76,47 \end{bmatrix} mm^2$$

Notice that with denomination millimeter instead of meter, the values are “easier” to read, without all the zeros.

The transformation matrix, (10.5) in the text book “Basic GPS”, entering the latitude and longitude of point C1-2:

$$T = \begin{bmatrix} -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \\ -\sin \lambda & \cos \lambda & 0 \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi \end{bmatrix} = \begin{bmatrix} -0,880804199 & -0,17943227 & 0,438164377 \\ -0,199614306 & 0,979874548 & 0 \\ 0,429346121 & 0,087463878 & 0,898894865 \end{bmatrix}$$

Thus the variance-covariance propagation from the geocentric to the local geodetic (topocentric) system can be derived directly from the error propagation law: $C_{LG} = T \cdot C_{CTRS} \cdot T^T$, denomination mm^2 .

$$C_{xyz} = C_{LG} = \begin{bmatrix} 37,6078 & -6,7165 & 10,4347 \\ -6,7165 & 13,7811 & -2,5371 \\ 10,4347 & -2,5371 & 75,5311 \end{bmatrix} mm^2$$

The local geodetic coordinate system is oriented with the north axis, x axis, in the meridian trough the point, and with the ellipsoidal normal as the z axis (ellipsoidal height). Therefore the transformation from LG to the UTM map projection plane and NN2000 height is a pure mathematical transformation, with no changes of the variance-covariance matrix:

$$C_{UTM-NE,H} = C_{LG}, \text{ see above for values}$$

Calculate the standard deviations of the UTM coordinates and the NN2000 height.

In mm, square root of the elements in the main diagonal in matrix $C_{UTM-NE,H}$:

$$s_N = \sqrt{37,6078} = 6,13 \text{ mm} \quad s_E = \sqrt{13,7811} = 3,71 \text{ mm} \quad s_H = \sqrt{75,5311} = 8,69 \text{ mm}$$

Calculate the three correlation coefficients between the UTM coordinates and the NN2000 height.

Formulas are found in “Basic GPS”, chapter 10.1.1.

The correlation coefficient between two variables, is calculated from the standard deviations of the two variables and the covariance between the two variables. Here, the calculation of the correlation coefficient between N and E is shown:

$$k_{ij} = \frac{\sigma_{ij}}{\sigma_i \cdot \sigma_j} = k_{NE} = \frac{\sigma_{NE}}{\sigma_N \cdot \sigma_E} = \frac{-6,7165}{\sqrt{37,6078} \cdot \sqrt{13,7811}} = -0,29503 \quad , \quad k_{ii} = 1 \quad , \quad -1 \leq k \leq 1$$

The correlation matrix of the UTM coordinates N and E , and the NN2000 height H :

$$\mathbf{K}_{NEH} = \begin{bmatrix} 1 & k_{NE} & k_{NH} \\ k_{EN} & 1 & k_{NH} \\ k_{HN} & k_{HN} & 1 \end{bmatrix} = \begin{bmatrix} 1 & -0,29503 & 0,19579 \\ -0,29503 & 1 & -0,07864 \\ 0,19579 & -0,07864 & 1 \end{bmatrix}$$

Check your results by comparing with the values from table 2: sX, sY, sH, and rXY, rXZ and rYZ.

sX	sY	sH	rXY	rXH	rYH
0.00445	0.003	0.010	-0.156	0.396	-0.165
0.00613	0.004	0.009	-0.295	0.196	-0.079

Exactly the same values! NB: Standard deviation: $\sigma = s$ in Gisline. Correlation coefficient: $k = r$ in Gisline

Gross error test. Grovfeilanalyse (observasjonstest)

Relevant theory in the text book “Reliability analyses of networks of control points”.

Parameters of the multiple T test, compare your calculations with values in the printout above:

Show that the degree of freedom, f , of the test is 11.

f = number of measurements – number of unknown coordinates x and y (2D) – the actual gross error

$$f = (3 + 3 + 2 + 2) \cdot 2 - 4 \cdot 2 - 1 = 20 - 8 - 1 = 11$$

Show that the confidence level, α , of the test is $0,0013 = 0,13\%$.

Only 20 observations. We have to calculate the confidence level (significance level) for 1 observation, instead of using the value $0,1\% = 0,001$ that is valid when there are more than 50 observations. The goal is to have a total confidence level of $5\% = 0,05$ for the multiple student T test. See the compendium “Reliability analyses of networks...”.

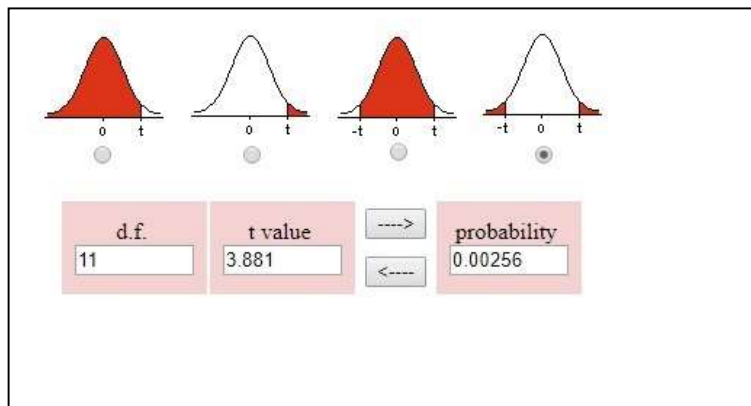
Formula:

$$\alpha_{TOT} = 1 - (1 - \alpha)^n, \text{ and: } \alpha = 1 - (1 - \alpha_{TOT})^{1/n} = 1 - (1 - 0,05)^{1/20} = 0,00256 = 0,256 \%$$

Normally we are using a 2-sided test, then the confidence level for 1 observation is: $\alpha = 0,256 / 2 = 0,128 \% = 0,0013$

Check that the tabulated value of the test is 3,88.

Using one of many Student's T-distribution calculators: <https://surfstat.anu.edu.au/surfstat-home/tables/t.php> (2018):



The tabulated value of the 2-sided $T_{f=11, \alpha=0,256\%} = 3,881$

Results of the gross error test:

Explain residual and estimated gross error in the report of the gross error test.

For more theory: See chapter B.1.1 in text book “Reliability analyses of networks of control points”.

UTJEVNING I GRUNNRIS (estimation in the plane, 2D)					
FRI UTJEVNING (free estimation, free network)					
TEST AV OBSERVASJONER - MULTIPPEL T-TEST					
Fra	Til		Restfeil	Est.grovfeil	Testverdi
			Residual	Estimated gross error	Test value
C1	1	X	-0.011	0.019	1.00
C1	1	Y	0.010	-0.014	1.12
C1	2	X	-0.015	0.028	1.37
C1	2	Y	-0.009	0.017	1.39
C1	3	X	0.063	-0.067	4.46 *
C1	3	Y	-0.012	-0.003	0.18

Residual (Restfeil): Corresponds to the correction the actual observation gets by the LSM estimation of the coordinates.

Example: The north coordinate of point C1, 3 coordinate observations (3 rounds):

	Punkt	Mnr		Tema	Inr	Type	X	Y	H	Ant.h				sX	sY	sH
	C1	1	{PUNKTOBS}		61	P	7100907.53200	622902.774	74.19462	0.000				0.00445	0.003	0.010
	C1	2	{PUNKTOBS}		61	P	7100907.53600	622902.793	74.22062	0.000				0.00613	0.004	0.009
*	C1	3	{PUNKTOBS}		61	P	7100907.4580	622902.796	74.1826	0.000				0.0078	0.004	0.014

Weighted mean value of the 3 N (X) coordinate observations of point C1:

Using the standard deviations sX to calculate the weights, and choosing observation no. 1 as the observation with weight equal $p_1 = 1$ in the equation of variances and weights $s_1^2 \cdot p_1 = 0,00445^2 \cdot 1 = s_2^2 \cdot p_2 = s_3^2 \cdot p_3$, $p_2 = 0,5270$, $p_3 = 0,3255$

Weighted mean: $x = 7100907,5201$ m

Residuals: $v_1 = x - l_1 = 7100907,5201 - 7100907,532 = -0,0119$ m, $v_2 = x - l_2 = -0,0159$ m, $v_3 = x - l_3 = 0,0621$ m
 From Gisline: $-0,011$ m, $-0,015$ m, $0,063$ m

Here the corrections are calculated, and they have the same sign as in Gisline. In the compendium the correction is defined to close the error, and the sign of the error is opposite to the sign of the correction.

So, using this definition, the residual/restfeil in Gisline is actually the estimated correction on the actual observation.

Estimated gross error (Estimert grovfeil): The largest gross error the actual observation can have, is calculated by a statistical test (is a statistical error), *the worst possible case*. The method is described in the compendium.

Comment the result of the gross error test. Are there any gross error(s)?

A gross error on the north coordinate in observation no. 3 for point C1: The test value of the statistical test of 4,46 is larger than the tabulated Student-T value of 3,88.

The test value is calculated from the observations:

$t = (\text{standard deviation of the observation} / \text{standard deviation of the corresponding gross error})$.

The max value, the tabulated Student-T value, is calculated from the number of excess observations, here $f = 11$, and the chosen value of the significance level, here 5 % which gives $\alpha = 0,00128 = 0,128$ %.

See the answers above for details.

Tabell 10.9 Observasjonstest

TEST AV OBSERVASJONER - MULTIPPEL T-TEST					
Fra	Til		Restfeil	Est.grovfeil	Testverdi
C1	1	X	-0.011	0.019	1.00
C1	1	Y	0.010	-0.014	1.12
C1	2	X	-0.015	0.028	1.37
C1	2	Y	-0.009	0.017	1.39
C1	3	X	0.063	-0.067	4.46 *
C1	3	Y	-0.012	-0.003	0.18
Tabellverdi=3.88 (Student-t, f=11, alfa=0.0013)					

The observations of the other 3 points (C2, C3, C4) passed the test, they have no gross errors. The test values were from 0,12 to 1,39, which is well inside the max value of the tabulated value of 3,88.

Probably you found a gross error. Describe the procedure after gross error(s) is/are detected.

Because of the gross error found in observation no. 3 of point C1, this observation is not used, only observations no. 1 and 2 are used in the internal and external reliability analyses.

To be sure about the result, a new gross error test was done without obs. no. 3 of C1. This extra test did not find any new gross error, see the results in the table below. It is important to carry out this extra test after one of the observation with gross error is removed.

New gross error test, without point observation C1-3, All the test values are within the tabulated Student-T value of 4,06. NB: We removed 2 observations, N and E of observation C1-3, then the degree of freedom is $f = 18 - 8 - 1 = 9$, and the corresponding new value of alpha is 0,0014:

TEST AV OBSERVASJONER					
DATUM: EUREF89 - SONE 32					
UTJEVNING I GRUNNRIS					
FRI UTJEVNING					
UTJEVNING I GRUNNRIS (estimation in the plane, 2D)					
FRI UTJEVNING (free estimation, free network)					
TEST AV OBSERVASJONER - MULTIPPEL T-TEST					
Fra	Til		Restfeil	Est.grovfeil	Testverdi
C1	1	X	0.002	-0.011	0.83
C1	1	Y	0.009	-0.020	3.18
C1	2	X	-0.002	0.011	0.83
C1	2	Y	-0.010	0.020	3.18
C2	1	X	0.004	-0.008	0.49
C2	1	Y	0.003	-0.005	0.45
C2	2	X	0.002	-0.003	0.20
C2	2	Y	0.002	-0.003	0.33
C2	3	X	-0.007	0.013	0.77
C2	3	Y	-0.002	0.006	0.70
C3	1	X	-0.002	0.015	0.69
C3	1	Y	-0.015	0.021	1.47
C3	2	X	0.006	-0.015	0.69
C3	2	Y	0.005	-0.021	1.47
C4	1	X	0.007	-0.011	0.61
C4	1	Y	-0.007	0.012	1.05
C4	2	X	-0.006	0.011	0.61
C4	2	Y	0.006	-0.012	1.05
Tabellverdi=4.06 (Student-t, f=9, alfa=0.0014)					
OPPSUMERING ETTER TEST AV OBSERVASJONER:					
Kategori		Ant.obs.	Akkumulert (%)		
Test/Tabell < 1.0		18	100.00		
1.0 < Test/Tabell < 2.0		0	100.00		
2.0 < Test/Tabell < 3.0		0	100.00		
3.0 < Test/Tabell < 3.0		0	100.00		
Ukontrollerbar		0	100.00		

External reliability. «Ytre pålitelighet»

The result of the external reliability test is shown below (is table 5):

YTRE PÅLITELIGHET		External reliability				
DATUM: EUREF89 - SONE 32						
UTJEVNING I GRUNNRIS		Estimation in 2D				
TVUNGEN UTJEVNING		Constraint estimation				
NYBESTEMTE KOORDINATER MED MIDLERE FEIL (standardavvik) [meter]						
Estimated coordinates and their standard deviations						
PUNKT (point)	N	E	H	sN	sE	sH
C1	7100907.534	622902.783		0.006	0.004	
C2	7100906.331	622899.662		0.008	0.004	
C3	7100876.016	622917.939		0.010	0.006	
C4	7100881.632	622933.828		0.008	0.006	
YTRE PÅLITELIGHET - KOORDINATER [meter]						
External reliability - Coordinates						
KOORDINAT	Observasjon.... [meter/gon]			Indre pål.	Ytre pål.	
Coordinate	Observation,no.			Internal reli.	External rel.	
P C1	C1	1	X	-0.041	0.027	
P C2	C2	1	X	-0.043	0.015	
P C3	C3	2	Y	-0.054	0.043	
P C4	C4	1	X	-0.050	0.031	
Normparametre:						
Normtype	Stedfesting av matrikkelenhets- og råderettsgrenser					
Norm type	Geo referencing of land register- and right-of-use borders					
Normklasse	1 : Byområde (høy grad av utnyttning) (k = 100 mm)					
Norm class	1 : Urban area, high degree of utilization (k = 100 mm)					
Tabellverdi = 2.26 (Student-t, f = 9, alfa = 0.0250)						
Tabulated value = 2.26 (Student-t, f = 9, alpha = 0.0250)						

Parameters of the external reliability test: compare your calculations with values in the printout above:

Show that the degree of freedom, f , of the test is 9.

f = number of measurements – number of unknown coordinates x and y (2D) – the actual gross error

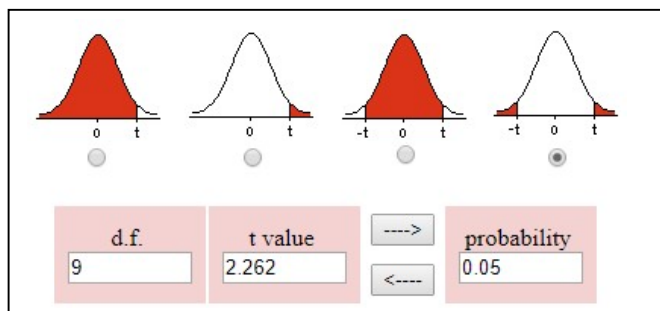
$$f = (2 + 3 + 2 + 2) \cdot 2 - 4 \cdot 2 - 1 = 18 - 8 - 1 = 9$$

Why is the confidence level (significance level), α , of the test $0,0250 = 2,5\%$.

Chosen significance level is 5 %, and a two sided test gives $\alpha = 0,025 = 2,5 \%$.

Check that the tabulated Student-T value of the test is 2,26.

Using one of many Student's T-distribution calculators: <https://surfstat.anu.edu.au/surfstat-home/tables/t.php> (2018):



The tabulated value of the 2-sided $T_{f=9, \alpha=5\%} = 2,262$

Explain the actual chosen “Norm class”

Using the Norwegian standard *Stedfesting av matrikkelenhets- og råderettsgrenser* («Geo referencing of land register- and right-of-use borders») The table below is copied from chapter 5.1.2. It shows the tolerances of the external reliability in various area types («områdetyper»):

Krav til ytre pålitelighet ved stedfesting av matrikkelenhetsgrenser (maksimalt tillatt punktdeformasjon)			
Områdetype (se kapittel 4)	Matrikkelenheter unntatt punktteste		Punktteste
	grunnriss	høyde, bare anleggseiendom	
1	0,10 m	0,10 m	
2	0,10 m	0,10 m	0,50 m
3a	0,10 m	0,10 m	0,50 m
3b	0,50 m	0,50 m	2,00 m
4	0,50 m	0,50 m	2,00 m

Tabell 1. Krav til ytre pålitelighet i stedfestingen av matrikkelenhetsgrenser.

Chosen norm class, area type (normklasse, områdetype) is area type no. 1, Urban area, high degree of development (Byområde, høy grad av utnyttning), with the corresponding parameter in the plane (“grunnriss”): $k = 0,10$ m.

Results of the external reliability test:

Why do we prefer to use a constraint network for external reliability analysis?

Constrained network, constrained estimation:

A test on external reliability is performed on a *constrained network*, i.e. by holding the control points which have known coordinates and heights fixed. By using a constrained network instead of a free network any remaining errors (or the accuracy of the known control points) also will affect the outcome of the external reliability. This effect is in many cases a desired effect.

In this example, is it a constrained network?

No, the observation method is *point observations*. The 4 points have no known coordinates, and there are no baselines measured between the 4 points, to tie them together.

Explain the values of the internal reliability (“indre nøyaktighet”) in the table above.

Internal reliability, is the largest remaining gross error. See chapter B.3.1 in the compendium “*Reliability analyses of networks of control points*”.

Comment the result of the external reliability test. Are the points Ok?

The four points were first estimated by use of least squares method (LSM), “minste kvadraters metode MKM”, see first in the table below. Thereafter internal and external reliability is done.

Table Test, external reliability (Gisline)

YTRE PÅLITELIGHET - KOORDINATER [meter]					
External reliability - Coordinates					
KOORDINAT	Observasjon....[meter/gon]			Indre pål.	Ytre pål.
Coordinate	Observation,no.			Internal reli.	External rel.

P C1	C1	1	X	-0.041	0.027
P C2	C2	1	X	-0.043	0.015
P C3	C3	2	Y	-0.054	0.043
P C4	C4	1	X	-0.050	0.031

Normparametre:

Normtype : Stedfesting av matrikkelenhets- og råderettsgrenser

Norm type : Geo referencing of land register- and right-of-use borders

Normklasse : 1 : Byområde (høy grad av utnyttning) (k = 100 mm)

Norm class : 1 : Urban area, high degree of utilization (k = 100 mm)

Tabellverdi = 2.26 (Student-t, f = 9, alfa = 0.0250)

Tabulated value = 2.26 (Student-t, f = 9, alpha = 0.0250)

All the four points fulfil the tolerances in the Norwegian standard “*Stedfesting av matrikkelenhets- og råderettsgrenser*” (“Geo referencing of land register- and right-of-use borders”): The values of the external reliability are smaller than the tolerances in the standard of 0,10 m, when we are using the area type *Urban area* (områdetypen *Byområde*), area with high degree of utilization.

Why does point C2 have a much lower numerical value of the external reliability compared to the other three points, 0,015 m vs 0,027/0,043/0,031 m?

Point C2 has the best result from the test, 0,015 m, compared to ca. 0,03-0,04 m for the other 3 points. The reason for this is: C2 is measured-in three times, and the others only two times. The more measurements, the more sure we are that the coordinates are good. An increased number of observations will generally give that the observations are better controlling each other.

For point C2: Explain in words what causes the external reliability value of 0,015 m.

Maximum deformation of point C2 is 0,015 m (sum of the deformations on both x (N) and y (E) directions). The reason for this maximum deformation is a possible remaining gross error (from the internal reliability) on the x (N) coordinate in observation no. 1 (C2-1) of -0,043 m..

Compare the external reliability values of points C1 and C2 with the standard deviations of the estimated coordinates shown in the table above.

YTRE PÅLITELIGHET	External reliability					
DATUM: EUREF89 - SONE 32						
UTJEVNING I GRUNNRIS	Estimation in 2D					
TVUNGEN UTJEVNING	Constraint estimation					
NYBESTEMTE KOORDINATER MED MIDLERE FEIL (standardavvik) [meter]						
Estimated coordinates and their standard deviations						
PUNKT (point)	N	E	H	sN	sE	sH

C1	7100907.534	622902.783		0.006	0.004	
C2	7100906.331	622899.662		0.008	0.004	
C3	7100876.016	622917.939		0.010	0.006	
C4	7100881.632	622933.828		0.008	0.006	

Notice that the standard deviations (sN, sE) of 0,006 and 0,004 m for C1 and 0,008 and 0,004 m for C2 in x (N) and y (E) directions are much smaller than the possible maximum point deformation.

The reason is that the tolerances of the external reliability represent the «worst case) of deformations, as we are using the largest possible gross errors in the reliability analysis.

For C1, C3 and C4 the differences between the tolerances and the standard deviations are larger. This is due to the different number of observations in each point for these three points; only two observations, compared to three observations of point C2.

Relative large remaining gross errors (from the internal reliability) is explained by the control of the observations in this case is not especially effective; there are only two or three observations of each point. With few extra observations, like here, the method will give relatively large values of the internal reliability; that is, large remaining gross errors can be hidden among the observations. There rather large statistically remaining gross errors will in the external reliability test cause large deformations.

Here, the result nevertheless is Ok, the external reliability values are within the tolerance of 0.10 m, given in the table above, on top of page 10.